



Smart Speaking Signs Device (SSSD)

Abstract

The Smart Speaking Signs Device (SSSD) is a wearable system that translates sign language into text and speech in real-time. It helps deaf and mute individuals communicate without interpreters. The system uses a smart glove with sensors, an ESP32 controller, and a Flutter app connected via Bluetooth and Firebase. SSSD is a low-cost, portable solution that improves accessibility in hospitals and daily life.

Methodology and System Design

Hardware Components:

- ❖ **ESP32 Microcontroller:** Processes sensor data and handles Bluetooth communication
- ❖ **Flex Sensors (x5):** Attached to each finger to measure bending and finger movement
- ❖ **MPU6050 (IMU):** Detects hand orientation and motion using accelerometer and gyroscope
- ❖ **Li-Ion Battery + TP4056 Charger:** Provides portable and rechargeable power supply for the glove

Software Components:

- ❖ **Arduino IDE:** Used to program the ESP32 for gesture recognition and BLE transmission
- ❖ **Flutter Mobile App:** Receives gesture codes, queries Firebase, and displays output
- ❖ **Firebase Realtime Database:** Stores mappings between gesture codes and corresponding phrases
- ❖ **Text-to-Speech (TTS):** Converts translated text into spoken audio through the mobile device

Wireless Communication:

- ❖ **Bluetooth Low Energy (BLE):** Enables real-time, low-power data transfer between the glove and the mobile application.

System Workflow:

- ❖ User performs a hand gesture.
- ❖ Flex and IMU sensors capture finger positions and hand movement.
- ❖ ESP32 processes the input and matches it to a gesture code.
- ❖ The code is sent via BLE to the mobile application.
- ❖ The app queries Firebase to retrieve the corresponding phrase.
- ❖ The phrase is displayed as text and spoken aloud using TTS.

Prototype

- Smart glove with 5 flex sensors and 1 IMU detects hand gestures.
- ESP32 processes data and sends gesture code via BLE.
- Flutter app receives code, fetches phrase from Firebase, and outputs text & speech.
- Rechargeable, portable system with <1s real-time response.
- Successfully tested in hospital settings.

Problem statement

Deaf and mute individuals face difficulty communicating in environments where sign language is not understood. Existing solutions are often expensive, complex, or lack real-time interaction—leading to isolation and limited access to essential services.

Objectives

- Design a smart glove to recognize hand gestures.
- Translate gestures into text and speech.
- Ensure real-time feedback (≤ 1 second).
- Promote independence and inclusion.
- Keep the system low-cost, usable, and portable.

Conclusion

The Smart Speaking Signs Device (SSSD) accurately translates hand gestures into text and speech in real-time. It is affordable, portable, and user-friendly—designed to assist deaf and mute individuals in daily communication, especially in critical settings like hospitals. The system lays a strong foundation for future upgrades, including multilingual and AI-powered features.

Students: Anwar Abu Bsharah ,
Masa Ayyad , Dina Abed

Supervisor: Dr. Allam AbuMwais